

TRYPHON T. GEORGIOU
Distinguished Professor, Mechanical and Aerospace Engineering
University of California, Irvine
ORCID: 0000-0003-0012-5447

Education:

- 1983 Doctor of Philosophy (Electrical Engineering), University of Florida, Gainesville, FL
- 1979 Diploma (Mechanical and Electrical Engineering), National Technical University of Athens, Greece

Position held:

- 2016–present Professor (Above Scale), Mechanical and Aerospace Eng., Univ. of California, Irvine, CA
- 2016–present Prof. Emeritus, Dept. of Electrical Eng. and Computer Science, Univ. of Minnesota, MN
- 1994–2016 Professor, Dept. of Electrical Eng. and Computer Science, Univ. of Minnesota, MN
- 1990–2016 Director, Center for Control Science and Dynamical Systems, University of Minnesota
- 1989–1994 Assoc. Prof., Dept. of Electrical Eng. and Computer Science, Univ. of Minnesota, MN
- 1988–1989 Associate Prof., Department of Electrical Engineering, Iowa State University, Ames, IA
- 1986–1988 Assistant Prof., Department of Electrical Engineering, Iowa State University, Ames, IA
- 1983–1986 Assistant Prof., Department of Electrical Engineering, Florida Atlantic University, FL

Endowed chairs and professorships:

- 2018–present Distinguished Professor, University of California, Irvine
- 2016–2018 Chancellor’s Professor, University of California, Irvine
- 2002–2016 Vincentine Hermes-Luh Chair (Electrical Engineering), University of Minnesota

Research Fields:

Stochastic Control, Geometry of Optimal Transport and Schrödinger Bridges, Analytic Interpolation and Moment Problems, Robust Control, Stochastic Thermodynamics, Mathematical Physics.

Academies:

- 2011 Foreign Member, Royal Swedish Academy of Engineering Sciences (IVA)

Fellowships:

- 2023 Fellow, Hellenic Institute for Advanced Studies (HIAS)
- 2022 Fellow, American Association for the Advancement of Science (AAAS)
- 2021 Fellow, Society for Industrial and Applied Mathematics (SIAM)
- 2016 Fellow, International Federation of Automatic Control (IFAC)
- 2000 Fellow, Institute of Electrical and Electronics Engineers (IEEE)

Notable awards:

- 1992–2017 IEEE Control Systems Society G.S. Axelby Outstanding Paper Award, received four times, for the years 1992, 1999, 2003, 2017

Professional Society Membership:

Society for Industrial and Applied Mathematics (SIAM)
Institute of Electrical and Electronics Engineers (IEEE)
American Mathematical Society (AMS)
American Physical Society (APS)
American Association for the Advancement of Science (AAAS)

Notable Professional Activities and Service:

- 2025 Spring Organizing committee: “Non-commutative Optimal Transport,” quarter-long program at IPAM, UCLA
- 2022 April Instructor: African Institute for Math. Sciences (AIMS), Quantum Leap Africa postdoctoral program, Rwanda
- 2021–2022 Member, Bode Award Committee, IEEE Control Systems Society
- 2018–2021 Chair, Awards, IEEE Control Systems Society
- 2016 July General Chair, 22nd International Symposium on the Mathematical Theory of Networks and Systems (MTNS 2016)
- 2015–2016 General Chair, “Thematic Year in Control,” Institute of Mathematics & its Applications, Univ. of Minnesota
- 2002–2005 Member, Board of Governors, IEEE Control Systems Society

Plenary presentations:

- 2000 14th International Symposium on the Mathematical Theory of Networks and Systems (MTNS 2000), Perpignan, France.
Title: *Analytic interpolation with degree constraint*
- 2007 46th IEEE International Conference on Decision and Control, New Orleans, Louisiana.
Title: *The Meaning of Distances in Spectral Analysis*
- 2024 22nd European Control Conference, Stockholm, Sweden.
Title: *Stochastic Control meets Non-equilibrium Thermodynamics: Fundamental limits of power generation in thermodynamic engines*
- 2025 25th International Conference on Control, Automation, and Systems, Incheon, Korea.
Title: *Schrödinger Bridges and Monge Cycles: From thermodynamics to ensemble control and back*

Editorial Service:

- 1988–1995 SIAM Journal on Control and Optimization
- 1991–1992 IEEE Transactions on Automatic Control
- 1995–2010 Systems and Control Letters
- 2008–2012 IMA Journal of Mathematical Control and Information
- 2008–2009 IEEE Transactions on Signal Processing
- 2024–present IEEE Control Systems Magazine (Book Review Editor)
- 2023–present Complex Analysis and Operator Theory, Springer

Edited volume:

Special Issue on Systems Theory, Control and Related Topics, dedicated to the memory of Allen Tannenbaum, Pure and Applied Functional Analysis, **11**(2), 2026 (with S. Reich and A. Zaslavski).

Research Funding:

Supported by NSF since 1987, by AFOSR since 1989, and by ARO since 2022.

Current awards:

- 2024–2027 AFOSR FA9550-24-1-0278, \$496,538: *Stochastic Control and Optimal Mass Transport: A Schrödinger Bridge Approach*
- 2024–2027 NSF ECCS-2347357, \$300,000: *Fundamentals of Power Generation from Thermal Anisotropy: A Stochastic Control Framework*
- 2022–2026 ARO W911NF-22-1-0292, \$267,698: *Geometric Network Theory with Applications to Heterogeneous Data Analysis*

PhD students:

2025 Jordi Ventura Siches (International Center for Numerical Methods in Engineering, Spain)
 2025 Asmaa Eldesoukey (Iowa State University)
 2023 Olga Movilla Miangolarra (Universidad de La Laguna, Spain)
 2023 Mahmoud Abdelgalil (University at Buffalo)
 2023 Anqi Dong (Royal Institute of Technology, Stockholm, Sweden)
 2022 Amirhossein Karimi (First American Financial Corp)
 2021 Zahra Askarzadeh (Meta)
 2020 Rui Fu (Nanjing University, China)
 2017 Hamza Farooq (University of Minnesota, Twin Cities)
 2016 Yongxin Chen (Georgia Tech, Atlanta)
 2013 Lipeng Ning (Harvard, Boston)
 2012 Xianhua Jiang (Starkey Technologies, Minnesota)
 2012 Shahrouz Mir Takyar (Uber Management, San Francisco)
 2007 Gerald Mtatifikolo (August Electronics, Alberta)
 2005 Ali Nasiri Amini (Google)
 2001 Subbarao Varigonda (Carrier Corp.)
 1997 Ian Fialho (Boeing)
 1993 Chin Chang (Semtech Corp., California)
 1992 Craig Shankwitz (Montana State University)

Postdoctoral advisees:

2013–14 Francesca Carli (University of Cambridge, UK)
 2015–16 Kaoru Yamamoto (Kyushu University)
 2016–17 Abhishek Halder (Iowa State University)
 2017–18 Luigia Ripani (Crédit Agricole Assurances, France)
 2019–21 Amir Taghvaei (University of Washington, Seattle)
 2019–22 Fariba Ariaei (UC Irvine and CSU Long Beach)
 2020–21 Rui Fu (Nanjing University, China)
 2023–24 Olga Movilla Miangolarra (Universidad de La Laguna, Spain)
 2025–26 Asmaa Eldesoukey (Iowa State University)
 2026–present Zachary Fritts (University of California, Irvine)

Selected distinctions of students and postdoctoral advisees:

Y. Chen	IEEE CSS Donald P. Eckman Award (2022), A.V. “Bal” Balakrishnan Award (2021), Simons–Berkeley Research Fellowship (2021), IFAC Manfred Thoma Medal (2026); plenary lectures at the 2023 American Control Conference and at MTNS 2024
O. Movilla Miangolarra	Ilya Prigogine Prize for Thermodynamics (2025); Marie Skłodowska-Curie Postdoctoral Fellowship (2024); doctoral thesis, <i>Stochastic Thermodynamic Treatment of Thermal Anisotropy</i> , published by Springer (2024)
A. Eldesoukey	doctoral thesis, <i>Inference and Collective Steering for Stochastic and Deterministic Systems</i> , to appear with Springer

Former students and postdoctoral advisees hold faculty appointments at Georgia Tech, Harvard, the University of Washington, Kyushu University, Iowa State University, the University at Buffalo, and Nanjing University; in industry, I.J. Fialho is a Senior Technical Fellow at Boeing.

Representative publications in subject areas

Stochastic Control, Optimal Mass Transport, and Schrödinger Bridges:

Positive contraction mappings for classical and quantum Schrödinger systems, *Journal of Mathematical Physics*, **56**: 033301, 2015 (with M. Pavon).

Optimal steering of a linear stochastic system to a final probability distribution, Parts I, II, and III: *IEEE Transactions on Automatic Control*, Part I: **61**(5): 1158–1169, 2016; Part II: **61**(5): 1170–1180, 2016; Part III: **63**(9): 3112–3118, 2018 (with Y. Chen and M. Pavon).

On the relation between optimal transport and Schrödinger bridges: A stochastic control viewpoint, *Journal of Optimization Theory and Applications*, **169**: 671–691, 2016 (with Y. Chen and M. Pavon).

The most likely evolution of diffusing and vanishing particles: Schrödinger bridges with unbalanced marginals, *SIAM Journal on Control and Optimization*, **60**(4): 2016–2039, 2022 (with Y. Chen and M. Pavon).

Collective steering in finite time: Controllability on $GL^+(n, \mathbb{R})$, *IEEE Transactions on Automatic Control*, **70**(11): 7554–7563, 2025 (with M. Abdelgalil).

The holonomy of optimal mass transport: The Gaussian-linear case, *IEEE Transactions on Automatic Control*, **71**(2): 931–945, 2026 (with M. Abdelgalil).

Robust Control, Analytic Function Theory, and Moment Problems:

A topological approach to Nevanlinna–Pick interpolation, *SIAM Journal on Mathematical Analysis*, **18**(5): 1248–1260, 1987.

On the computation of the gap metric, *Systems and Control Letters*, **11**(4): 253–257, 1988.

Optimal robustness in the gap metric, *IEEE Transactions on Automatic Control*, **35**(6): 673–686, 1990 (with M.C. Smith).

Robust stability of feedback systems: A geometric approach using the gap metric, *SIAM Journal on Control and Optimization*, **31**(6): 1518–1537, 1993 (with C. Foias and M.C. Smith).

Generalized interpolation in H^∞ with a complexity constraint, *Transactions of the American Mathematical Society*, **358**(3): 965–987, 2006 (with C.I. Byrnes, A. Lindquist, and A. Megretski).

Relative entropy and the multivariable multidimensional moment problem, *IEEE Transactions on Information Theory*, **52**(3): 1052–1066, 2006.

Stochastic Thermodynamics and Mathematical Physics:

Colour of turbulence, *Journal of Fluid Mechanics*, **812**: 636–680, 2017 (with A. Zare and M.R. Jovanović).

Energy harvesting from anisotropic fluctuations, *Physical Review E*, **104**(4): 044101, 2021 (with O.M. Miangolarra, A. Taghvaei, R. Fu, and Y. Chen).

On the relation between information and power in stochastic thermodynamic engines, *IEEE Control Systems Letters*, **6**: 434–439, 2022 (with A. Taghvaei, O.M. Miangolarra, R. Fu, and Y. Chen).

Surveys and expository articles

Linear models based on noisy data and the Frisch scheme, *SIAM Review*, **57**(2): 167–197, 2015 (with L. Ning, A. Tannenbaum, and S.P. Boyd).

Stochastic control liaisons: Richard Sinkhorn meets Gaspard Monge on a Schrödinger bridge, *SIAM Review*, **63**(2): 249–313, 2021 (with Y. Chen and M. Pavon).

Optimal transport in systems and control, *Annual Review of Control, Robotics, and Autonomous Systems*, **4**: 89–113, 2021 (with Y. Chen and M. Pavon).

Stochastic dynamical modeling of turbulent flows, *Annual Review of Control, Robotics, and Autonomous Systems*, **3**: 195–219, 2020 (with A. Zare and M.R. Jovanović).

Optimal survival strategies for diffusive flows: A Schrödinger bridge approach to unbalanced transport, *SIAM Review*, **67**(3): 579–604, 2025 (with Y. Chen and M. Pavon).

Journal articles

J-171. M. Abdelgalil, **T.T. Georgiou**, Control of the State Transition Matrix via Factorization Into Positive Definite Factors, *IEEE Control Systems Letters*, **10**: 2083–2088, 2026, doi 10.1109/LCSYS.2026.3715083

J-170. R. Sabbagh, A. Eldesoukey, M. Abdelgalil, and **T.T. Georgiou**, Minimizing control attention: The linear Gauss–Markov paradigm, arxiv.org/abs/2512.07046, *IEEE Transactions on Automatic Control*, accepted for publication.

J-169. W. Wu, J. Chen, M.R. Jovanović, **T.T. Georgiou**, “Tannenbaum’s gain-margin optimization meets Polyak’s heavy-ball algorithm,” *IEEE Transactions on Automatic Control*, **71**(9): 5817–5832, Sept. 2026, doi 10.1109/TAC.2026.3678501

J-168. K. Xu, A. Bhardwaj, S. Koundal, Q. Ren, C. You, X. Papademetris, H. Benveniste, and **T. T. Georgiou**, Chronic hypertension impairs lymphatic drainage in deep cervical lymph nodes, *JCI Insight*, 2026, doi 10.1172/jci.insight.206714

J-167. L.D. Abreu, D. Alpay, **T.T. Georgiou**, P. Jorgensen, Analytic continuation of time in Brownian motion. Stochastic distributions approach, *Journal of Mathematical Analysis and Applications*, **558**(1), 130438 (18 pages), 2026, doi 10.1016/j.jmaa.2026.130438

J-166. Y. Chen, **T.T. Georgiou**, M. Pavon, Control and estimation of multi-commodity network flow under aggregation, *Pure and Applied Functional Analysis*, **11**(2): 251–266, 2026, <http://yokohamapublishers.jp/online2/pafav11n2.html>

J-165. A. Dong, A. Taghvaei, and **T.T. Georgiou**, A Lasso-alternative to Dijkstra’s algorithm for identifying short paths in networks, *Networks*, **88**(1): 28–41, 2026, doi 10.1002/net.70035

J-164. D. Grange, Y. Sun, C. Weistuch, R. Elkin, S. Phillips, J.O. Deasy and **T.T. Georgiou**, Curvature on Graphs With Negative Edge Weights, *IEEE Transactions on Network Science and Engineering*, **13**: 9205–9222, 2026, doi 10.1109/TNSE.2026.3682565

J-163. R. Sabbagh, A. Eldesoukey, M. Abdelgalil, and **T.T. Georgiou**, On the isospectral nature of minimum-shear covariance control, *IEEE Control Systems Letters*, **10**: 1171–1176, 2026, doi 10.1109/LC-SYS.2026.3706320.

- J-162. H. Farooq, Y. Chen, G. Rasool, **T.T. Georgiou**, and C. Lenglet, Diffusion MRI Experimental Design Optimization for Microstructure Imaging, *Communications Biology*, 2026, doi 10.1038/s42003-026-10256-2
- J-161. M. Abdelgalil, **T.T. Georgiou**, The holonomy of optimal mass transport: The Gaussian-linear case, *IEEE Transactions on Automatic Control*, **71**(2): 931–945, Feb. 2026, doi 10.1109/TAC.2025.3598675
- J-160. M. Abdelgalil, **T.T. Georgiou**, Collective steering in finite time: Controllability on $GL^+(n, \mathbb{R})$, *IEEE Transactions on Automatic Control*, **70**(11): 7554–7563, Nov. 2025, doi 10.1109/TAC.2025.3574186
- J-159. S.Z. Khong, L. Su, and **T.T. Georgiou**, Long-term susceptible fractions in networked epidemic models and their relation to the basic reproduction number, *Scientific Reports*, **15**: 38429, 2025, doi 10.1038/s41598-025-22158-0
- J-158. Yongxin Chen, **T.T. Georgiou**, and Michele Pavon, Optimal Survival Strategies for Diffusive Flows: A Schrödinger Bridge Approach to Unbalanced Transport, *SIAM Review*, **67**(3): 579–604, 2025, doi 10.1137/25M176581X
- J-157. Olga Movilla Miangolarra, Ralph Sabbagh, **T.T. Georgiou**, Quantum Schrödinger bridges: Large deviations and time-symmetric ensembles, *Physical Review A*, **112**(1): 012202, 2025, doi 10.1103/k35b-rkct
- J-156. O. Movilla Miangolarra, A. Eldesoukey, A. Movilla Miangolarra, and **T.T. Georgiou**, Maximum entropy inference of reaction-diffusion models, *Journal of Chemical Physics*, **162**(19), 2025, doi 10.1063/5.0256659
- J-155. Ralph Sabbagh, Olga Movilla Miangolarra, Hamid Hezari, **T.T. Georgiou**, Particle approximations of Wigner distributions for n arbitrary observables, *Physical Review Research*, **7**(1): 013102, 2025, doi 10.1103/PhysRevResearch.7.013102
- J-154. A. Eldesoukey and **T.T. Georgiou**, “Schrödinger’s control and estimation paradigm with spatio-temporal distributions on graphs,” *IEEE Transactions on Automatic Control*, **70**(4): 2466–2478, 2025, doi 10.1109/TAC.2024.3485537
- J-153. A. Eldesoukey, O. Movilla Miangolarra, and **T.T. Georgiou**, Inferring evanescent stochastic dynamics from marginal distributions, *Journal of Systems Science and Complexity*, **38**(1): 129–149, 2025, doi 10.1007/s11424-025-4440-9
- J-152. Q. Mao, Y. Xu, J. Chen, J. Chen, and **T.T. Georgiou**, Maximization of gain/phase margins by PID control, *IEEE Transactions on Automatic Control*, **70**(1): 34–49, January 2025, doi 10.1109/TAC.2024.3417717
- J-151. Anqi Dong, **T.T. Georgiou**, Allen Tannenbaum, Data Assimilation for Sign-indefinite Priors: A generalization of Sinkhorn’s algorithm, *Automatica*, **177**: 112283, 2025, doi 10.1016/j.automatica.2025.112283
- J-150. Arthur Stephanovitch, Anqi Dong, **T.T. Georgiou**, Optimal transport through a toll station, *European Journal of Applied Mathematics*, **36**(3): 613–637, 2025, doi 10.1017/S0956792524000317
- J-149. O. Movilla Miangolarra, A. Taghvaei, and **T.T. Georgiou**, Minimal entropy production in the presence of anisotropic fluctuations, *IEEE Transactions on Automatic Control*, **70**(2): 845–860, 2025, doi 10.1109/TAC.2024.3436748
- J-148. R. Fu, O. Movilla Miangolarra, A. Taghvaei, Y. Chen, and **T.T. Georgiou**, Stochastic thermodynamic engines under time-varying temperature profile, *Automatica*, **159**: 111361, 2024, doi 10.1016/j.automatica.2023.111361
- J-147. Anqi Dong, Arthur Stephanovitch, **T.T. Georgiou**, Monge–Kantorovich optimal transport through constrictions and flow-rate constraints, *Automatica*, **160**: 111448, 2024, doi 10.1016/j.automatica.2023.111448

- J-146. S. Zhang, W. Wu, Z. Li, J. Chen and **T.T. Georgiou**, “Frequency-domain analysis of distributed optimization: Fundamental convergence rate and optimal algorithm synthesis,” *IEEE Transactions on Automatic Control*, **69**(12): 8539–8554, December 2024, doi 10.1109/TAC.2024.3413854
- J-145. A. Eldesoukey, O. Movilla Miangolarra and **T.T. Georgiou**, “An excursion onto Schrödinger’s bridges: Stochastic flows with spatio-temporal marginals,” *IEEE Control Systems Letters*, **8**: 1138–1143, 2024, doi 10.1109/LCSYS.2024.3409107
- J-144. J. Ventura Siches, O. Movilla Miangolarra, and **T.T. Georgiou**, “Refined bounds on energy harvesting from anisotropic fluctuations,” *Physical Review E*, **109**(6): 064155, 2024.
- J-143. R. Sabbagh, O. Movilla Miangolarra, and **T.T. Georgiou**, Wasserstein speed limits for Langevin systems, *Physical Review Research*, **6**(3): 033308, September 2024, doi 10.1103/PhysRevResearch.6.033308
- J-142. O. Movilla Miangolarra, A. Eldesoukey, and **T.T. Georgiou**, “Inferring potential landscapes: A Schrödinger bridge approach to maximum caliber,” *Physical Review Research*, **6**(3): 033070, 2024.
- J-141. D. Grange, M. Al-Jarrah, R. Baptista, A. Taghvaei, **T.T. Georgiou** and A. Tannenbaum, Computational optimal transport and filtering on Riemannian manifolds, *IEEE Control Systems Letters*, **7**: 3495–3500, November 2023, doi 10.1109/LCSYS.2023.3331834
- J-140. Q. Mao, Y. Xu, J. Chen, **T.T. Georgiou**, Implementation-Oriented Filtered PID Control: Optimization of Robustness Margins, *Automatica*, **152**: 110974, 2023, doi.org/10.1016/j.automatica.2023.110974
- J-139. O. Movilla Miangolarra, A. Taghvaei, **T.T. Georgiou**, A matching principle for power transfer in stochastic thermodynamics, *IEEE Control Systems Letters*, **7**: 2107–2112, 2023, doi 10.1109/LCSYS.2023.3285544
- J-138. J. Ventura Siches, O. Movilla Miangolarra, A. Taghvaei, Y. Chen, **T.T. Georgiou**, Inertialess gyrating engines, *PNAS Nexus*, **1**(5): pgac251, 2022, doi 10.1093/pnasnexus/pgac251
- J-137. O. Movilla Miangolarra, A. Taghvaei, Y. Chen, **T.T. Georgiou**, Geometry of finite-time thermodynamic cycles with anisotropic thermal fluctuations, *IEEE Control Systems Letters*, **6**: 3409–3414, 2022, doi 10.1109/LCSYS.2022.3184912
- J-136. O. Movilla Miangolarra, A. Taghvaei, Y. Chen, **T.T. Georgiou**, Thermodynamic engine powered by anisotropic fluctuations, *Physical Review Research*, **4**: 023218, 2022, doi 10.1103/PhysRevResearch.4.023218
- J-135. F. Ariaei, Z. Askarzadeh, Y. Chen, **T.T. Georgiou**, Macroscopic network circulation for planar graphs, *IEEE Transactions on Control of Network Systems*, **9**(4): 1840–1850, 2022, doi 10.1109/TCNS.2022.3175127
- J-134. Y. Chen, **T.T. Georgiou**, and M. Pavon, The most likely evolution of diffusing and vanishing particles: Schrödinger bridges with unbalanced marginals, *SIAM Journal on Control and Optimization*, **60**(4): 2016–2039, 2022, doi 10.1137/21M1447672
- J-133. A. Taghvaei, O. M. Miangolarra, R. Fu, Y. Chen, **T.T. Georgiou**, “On the relation between information and power in stochastic thermodynamic engines,” *IEEE Control Systems Letters*, **6**: 434–439, 2022, doi 10.1109/LCSYS.2021.3078716
- J-132. O.M. Miangolarra, A. Taghvaei, R. Fu, Y. Chen, **T.T. Georgiou**, Energy harvesting from anisotropic fluctuations, *Physical Review E*, **104**(4): 044101, 2021, doi 10.1103/PhysRevE.104.044101
- J-131. O.M. Miangolarra, R. Fu, A. Taghvaei, Y. Chen, and **T.T. Georgiou**, Underdamped stochastic thermodynamic engines in contact with a heat bath with arbitrary temperature profile, *Physical Review E*, **103**(6): 062103, 2021, doi 10.1103/PhysRevE.103.062103

- J-130. Y. Chen, **T.T. Georgiou**, M. Pavon, Optimal Transport in Systems and Control, *Annual Review of Control, Robotics, and Autonomous Systems*, **4**: 89–113, 2021, doi 10.1146/annurev-control-070220-100858
- J-129. Y. Chen, **T.T. Georgiou**, M. Pavon, Stochastic control liaisons: Richard Sinkhorn meets Gaspard Monge on a Schrödinger bridge, *SIAM Review*, **63**(2): 249–313, 2021, doi 10.1137/20M1339982
- J-128. **T.T. Georgiou** and A. Lindquist, On a Fejér-Riesz factorization of generalized trigonometric polynomials, *Communications in Information and Systems*, **21**(3): 371–384, 2021, doi 10.4310/CIS.2021.v21.n3.a3
- J-127. R. Fu, A. Taghvaei, Y. Chen, **T.T. Georgiou**, Maximal power output of a stochastic thermodynamic engine, *Automatica*, **123**: 109366, 2021, doi 10.1016/j.automatica.2020.109366
- J-126. V. Ciccone, Y. Chen, **T.T. Georgiou** and M. Pavon, “Regularized transport between singular covariance matrices,” *IEEE Transactions on Automatic Control*, **66**(7): 3339–3346, 2021, doi 10.1109/TAC.2020.3017714
- J-125. Y. Chen, **T.T. Georgiou**, and M. Pavon, Controlling Uncertainty: Schrödinger’s Inference Method and the Optimal Steering of Probability Distributions, *IEEE Control Systems Magazine*, **41**(4): 82–94, 2021.
- J-124. A. Karimi, L. Ripani, and **T.T. Georgiou**, Statistical learning in Wasserstein space, *IEEE Control Systems Letters*, **5**(3): 899–904, July 2021, doi: 10.1109/LCSYS.2020.3006965
- J-123. A. Taghvaei, **T.T. Georgiou**, L. Norton, A.R. Tannenbaum, Fractional SIR epidemiological models, *Scientific Reports*, **10**: 20882, 2020, doi 10.1038/s41598-020-77849-7
- J-122. C. Guilloteau, A. Mamalakis, L. Vulis, **T.T. Georgiou**, and E. Foufoula-Georgiou, Rotated spectral principal component analysis (rsPCA) for identifying dynamical modes of variability in climate systems, *Journal of Climate*, 715–736, 2020, doi.org/10.1175/JCLI-D-20-0266.1
- J-121. A. Zare, H. Mohammadi, N.K. Dhingra, **T.T. Georgiou**, M.R. Jovanović, Proximal algorithms for large-scale statistical modeling and sensor/actuator selection, *IEEE Transactions on Automatic Control*, **65**(8): 3441–3456, August 2020, doi 10.1109/TAC.2019.2948268
- J-120. Y. Chen, **T.T. Georgiou**, and A. Tannenbaum, Stochastic control and non-equilibrium thermodynamics: fundamental limits, *IEEE Transactions on Automatic Control*, **65**(7): 2979–2991, July 2020, doi 10.1109/TAC.2019.2939625
- J-119. **T.T. Georgiou**, F. Jabbari, M.C. Smith, Principles of lossless adjustable one-ports, *IEEE Transactions on Automatic Control*, **65**(1): 252–262, January 2020, doi 10.1109/TAC.2019.2917853
- J-118. Y. Chen, **T.T. Georgiou**, M. Pavon, A. Tannenbaum, Relaxed Schrödinger bridges and robust network routing, *IEEE Transactions on Control of Network Systems*, **7**(2): 923–931, June 2020, doi 10.1109/TCNS.2019.2935623
- J-117. Z. Askarzadeh, R. Fu, A. Halder, Y. Chen, and **T.T. Georgiou**, Stability Theory of Stochastic Models in Opinion Dynamics, *IEEE Transactions on Automatic Control*, **65**(2): 522–533, Feb. 2020, doi 10.1109/TAC.2019.2912490
- J-116. H. Farooq, Y. Chen, **T.T. Georgiou**, A. Tannenbaum, C. Lenglet, Network Curvature as a Hallmark of Brain Structural Connectivity, *Nature Communications*, **10**, 4937 (2019) doi:10.1038/s41467-019-12915-x
- J-115. A. Zare, **T.T. Georgiou** and M. R. Jovanović, Stochastic dynamical modeling of turbulent flows, *Annual Review of Control, Robotics, and Autonomous Systems*, **3**: 195–219, 2020.
- J-114. Y. Chen, W. Gangbo, **T.T. Georgiou**, A. Tannenbaum, On the Matrix Monge-Kantorovich Problem, *European Journal of Applied Mathematics*, 2019, doi:10.1017/S0956792519000172.

- J-113. Y. Chen, **T.T. Georgiou**, A. Tannenbaum, Optimal transport for Gaussian mixture models, *IEEE Access*, **7**: 6269–6278, 2019, doi 10.1109/ACCESS.2018.2889838
- J-112. **T.T. Georgiou** and A. Lindquist, Dynamic Relations in Sampled Processes, *IEEE Control Systems Letters*, **3**(1): 144–149, January 2019, doi: 10.1109/LCSYS.2018.2859481
- J-111. Y. Chen, **T.T. Georgiou**, and M. Pavon, Steering the Distribution of Agents in Mean-Field Games, *Journal of Optimization Theory and Applications*, **179**(1): 332–357, 2018, doi.org/10.1007/s10957-018-1365-7
- J-110. Y. Chen, **T.T. Georgiou**, and A. Tannenbaum, Vector-Valued Optimal Mass Transport, *SIAM Journal on Applied Mathematics*, **78**(3): 1682–1696, 2018, doi.org/10.1137/17M1130897
- J-109. Y. Chen, **T.T. Georgiou**, A. Tannenbaum, Interpolation of matrices and matrix-valued densities: The unbalanced case, *European Journal of Applied Mathematics*, 1–23, May 2018, doi.org/10.1017/S0956792518000219
- J-108. Y. Chen, G. Conforti, **T.T. Georgiou**, Measure-valued spline curves: An optimal transport viewpoint, *SIAM Journal on Mathematical Analysis*, **50**(6): 5947–5968, 2018, doi.org/10.1137/18M1166249
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